

Matías Arim

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Research focus: the interaction between landscape structure, energy, and biodiversity, as determinants of metacommunity organization and function. I particularly attempt to connect theoretical and empirical studies.

A. Education:

The University of República, Uruguay	1999	B. Sc.	Biology
The University of República, Uruguay	2001	M.Sc.	Ecology
P. Catholic University of Chile, Chile	2005	PhD	Ecology
Center for Advanced Studies in Ecology and Biodiversity, Chile	2006	Postdoc	Ecology

B. Appointments:

2015-Present	Full Professor, Program for the Development of Basic Science, Uruguay
2005-2009	Assistant Professor, University of República, Uruguay
2009-Present	Associate Professor, University of República, Uruguay
2007-2011	CoPI, Center for Advanced Studies in Ecology and Biodiversity, CASEB, Chile (P. Catholic University of Chile).
2014-2018	Member of the Global Young Academy

C. Publications *Highlighted from 4 books chapters and 50 papers ISI*

1. Arim, M. & O. Barbosa. 2002. **Humped pattern of diversity: fact or artifact?** *Science*. 297:1763. <http://www.sciencemag.org/cgi/reprint/297/5588/1763a.pdf>
2. Arim, M. & P.A. Marquet. 2004. **Intraguild Predation: a significant and widespread interaction.** *Ecology Letters* 7: 557-564.
3. Arim M. & F. M. Jaksic. 2005. **Productivity and food web structure: association between productivity and link richness among top predators.** *Journal of Animal Ecology* 74: 31-40.
4. ¹Arim, M, S.R. Abades, P.E. Neill, M. Lima & P.A. Marquet. 2006. **Spread dynamics of invasive species.** *Proceeding of the National Academy of Science USA*. PNAS 103: 374-378.
5. Arim, M, Pablo A. Marquet & F.M. Jaksic. 2007. **On the relationship between productivity and food chain length at different ecological levels.** *The American Naturalist* 169: 62-72.
6. Arim, M, F. Bozinovic & Pablo A. Marquet. 2007. **On the relationship between trophic position, body mass and temperature: Reformulating the energy limitation hypothesis.** *OIKOS*. 116: 1524-1530.

1 Articles recommended in "Faculty of 1000".

7. Lafferty, K.D.; Allesina S.; **M. arim**; Briggs, C.J.; De Leo, G.; Dobson, A.P.; Dunne, J.A.; Johnson, P.T.J.; Kuris, A.M.; Marcogliese, D.J.; Martinez, N.D.; J. Memmott, J.; P.A. Marquet; J.P. Mclaughlin; E.A. Mordecai; M. Pascual; R. Poulin; D.W. Thieltges, D.W. 2008. **Parasites in food webs: the ultimate missing links.** *Ecology Letters* 11: 533-546.
8. Ramos-Jiliberto, R., A. Albornoz, F. Valdovinos, C. Smith-Ramírez, **M. Arim**, J. Armesto, P.A. Marquet. 2009. **Mutualistic networks, pollination web, degree distribution, nestedness, network dynamics.** *Oecologia*. 160: 697-706.
9. **Arim, M.**, S.R. Abades, G. Laufer, M. Loureiro & P.A. Marquet. 2010 **Food web structure and body size: trophic position and resource acquisition.** *Oikos* 119:147-153.
10. Toranza, C. & **M. Arim**. 2010. Cross-taxon congruence and environmental conditions. *BMC Ecology* 10:18.
11. Carranza, A. Defeo, O. & **Arim, M.** 2011. **Taxonomic relatedness and spatial structure of a shelf benthic gastropod assemblage.** *Diversity and Distribution* 17: 25-34.
12. **Arim, M.**, M. Berazategui, J. M. Barreneche, L. Ziegler, M. Zaruki, and S. R. Abades. 2011. **Determinants of density–body size scaling within food webs and tools for their detection.** *Advances in Ecological Research* 45:1-40.
13. Borthagaray, A.I., **Arim, M.** and Marquet, P.M. 2012. **Connecting landscape structure and patterns in body size distribution.** *Oikos* DOI: 10.1111/j.1600-0706.2011.19548.x).
14. Garay-Narvaez, L., **Arim, M.**, Flores, J.D., Ramos-Jiliberto, R. 2013. **The more polluted the environment, the more important biodiversity is for food web stability.** *Oikos*, 122: 1247-1253.
15. Ziegler L., Berazategui M. & **Arim M.** 2014 **Discontinuities and alternative scalings in the density–mass relationship of anuran larvae.** *Hydrobiologia*. 723:123-129. doi: 10.1007/s10750-013- 1553-2.
16. Borthagaray, A.I., Barreneche, J.M. Abades, S.R., & **Arim, M.** 2014 **Modularity along organism dispersal gradients challenges a prevailing view of abrupt transitions in animal landscape perception.** *Ecography*. 37(6):564-571. DOI: 10.1111/j.1600-0587.2013.00366.x
17. Garay-Narvaez, L., Flores, J.D., **Arim, M.**, Ramos-Jiliberto, R. 2014. **Food web modularity and biodiversity promote species persistence in polluted environments.** *Oikos*, 123: 583-588. DOI: 10.1111/j.1600-0706.2013.00764.x
18. Borthagaray, A.I., **Arim, M.**, Marquet, P.A. 2014. **Inferring species roles in metacommunity structure from species co-occurrence networks.** *Proceedings of the Royal Society B*. 281:. DOI: <http://dx.doi.org/10.1098/rspb.2014.1425>
19. Segura, A., Calliari, D. Kruk, C., Fort, H., Izaguirre, I., Saad, J., **Arim, M.** 2015 **Metabolic dependence of phytoplankton species richness.** *Global Ecology and Biogeography* 24: 472-482. DOI: 10.1111/geb.12258
20. Borthagaray, A.I., Berazategui, M., & **Arim, M.** 2015. **Disentangling the effects of local and regional processes on biodiversity patterns through taxon-contingent metacommunity network analysis.** *Oikos*. 124: 1383-1390. DOI: 10.1111/oik.01317
21. Borthagaray, A.I., Pinelli, V., Berazategui, M., Rodríguez-Tricott, L. and **M. Arim**. 2015. **Effects of metacommunity network on local communities structure: from theoretical predictions to empirical evaluations.** En *Aquatic Functional Biodiversity* A. Belgrano, G. Woodward and U. Jacob (Eds). Elsevier
22. **Arim, M.**, Borthagaray, A.I., Giacomini, H.C. 2016. **Energetic constrains to food chain lenght in a metacommunity framework.** *Candaina Journal of Fisheries and Aquatic Sciences*. 73(4): 685-692. DOI: 10.1139/cjfas-2015-0156
23. Segura, A. M., Fariña, R. A., & **Arim, M.** 2016. **Exceptional body sizes but typical**

- trophic structure in a Pleistocene food web.** *Biology Letters*, 12(5), 20160228
24. Segura, A. M., Calliari, D., Lan, B. L., Fort, H., Widdicombe, C. E., Harmer, R., & Arim, M. 2017. **Community fluctuations and local extinction in a planktonic food web.** *Ecology Letters*, 20(4), 471-476.
 25. Canavero, A., Arim, M., Pérez, F., Jaksic, F. M., & Marquet, P. A. (2018). **A metabolic view of amphibian local community structure: the role of activation energy.** *Ecography*.41: 388-400.
 26. Borthagaray, A.I., Soutullo, A., Carranza, A. Arim, M. 2018. A modularity-based approach for identifying biodiversity management units. *Revista Chilena de Historia Natural*. 91: 2.

D. Synergistic Activities:

D.1. Reviewer for Journals:

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| 1. Acta Oecologica | 20. Frontiers in Ecology |
| 2. Aquatic Ecology | 21. Functional Ecology |
| 3. Biological Invasions | 22. F1000Research |
| 4. Biological Research | 23. Gayana |
| 5. BMC Ecology | 24. Global Ecology and Biography |
| 6. Boletín de la Sociedad Zoológica del Uruguay | 25. Hydrobiologia |
| 7. Canadian Journal of Fisheries and Aquatic Sciences | 26. Ibis |
| 8. Canadian J. of Zoology | 27. Journal of Animal Ecology |
| 9. Climate Research | 28. Journal of Arid Environments |
| 10. Conservation Biology | 29. Journal of Biogeography |
| 11. Diversity and Distribution | 30. Journal of Ecology |
| 12. Ecography | 31. Journal of Mammalogy |
| 13. Ecological Entomology | 32. Journal of Zoology |
| 14. Ecological Modelling | 33. Marine Ecology Progress Series |
| 15. Ecology | 34. Marine Biodiversity |
| 16. Ecology Letters | 35. Neotropical Ichthyology |
| 17. Ecology and Evolution | 36. Oecologia |
| 18. Ecosistemas | 37. Oikos |
| 19. Evolutionary Ecology Research | 38. Revista Chilena de Historia Natural |
| | 39. Theoretical Ecology |
| | 40. Wildlife Research |

D.2. Reviewer of proposals

National Science Organization of Argentina, Austria, Chile, Colombia, Panamá, Uruguay, and for the European Research Council.

E. Advisor

Advisor of 16 undergraduate, 16 master, and 8 PhD students in programs of Argentina, Chile, Spain, and Uruguay. Now advisor of 3 Master, and 4 PhD students.

F. Principal Investigator in Grants *(highlighted from a total of 10)*

Arim, M. 2008-2018. Programs for University Development. PI of the Program "Development of Research and Human Resources on Biodiversity"

Arim, M. 2013-2015. "Metabolism, randomness, scale, and entropy: empirical evaluation of new theories in community Ecology" (Fondo Clemente Estable 2011, Metabolismo, azar, escalas y entropía: evaluación empírica de nuevas teorías en ecología de comunidades).

Arim, M. 2016-2018. "Effect of metacommunity network on local communities: from theoretical predictions to empirical evaluations" (Fondo Clemente Estable 2014, Efecto de la red metacomunitaria en las comunidades locales: desde las predicciones teóricas a la evaluación empírica). 2016-2017.

G. Responsible of undergraduate and graduate courses

Theory and methods in metacommunity ecology. Course directed to graduate students, organized with Dr. Ana Borthagaray in Argentina and Uruguay (since 2017).

Community Ecology. Course directed to graduate students (since 2012).

Ecology. Course directed to undergraduate students (since 2013).

Theoretical Ecology. Course directed to undergraduate students (since 2013).

F. Presentations

1. **2018. Arim M., Pinelli V., Rodríguez L., Ortiz E., Illarze M., Fagúndez C., Borthagaray A. Ponds size and isolation determining the relative role of niche versus neutral assembly of plants communities.** Nordic Society OIKOS, Trondheim, Norway.

2. **2017. Arim, M. Gastropods in food webs from general predictions to idiosyncratic patterns.** X Latin-American Congress of Malacology, Maldonado, Uruguay. Plenary Conference.

3. **2016. Arim, M. Metacommunities and Ecological Theory Integration.** Network for Ecological Theory Integration (NETI), Satan Fe Institute, USA.

4. **2015. Arim, M. The importance of take a look outside the pond to understand pond ecology.** F. de Ciencias, Universidad de la República, Uruguay. II Simposio de Peces Anuales. Plenary Conference.

5. **2015. Arim, M. Networks impacting networks: the interrelationship between metacommunities and food webs.** XXII Reunión Anual de la Sociedad de Ecología de Chile. Plenary Conference.